Pacific Stock Assessment
Review Committee

Working Paper S97-1
April 1997

APPROVED

Assessing the historical estimates of northern diversion of Fraser River sockeye salmon

by

Skip McKinnell
Fisheries and Oceans Canada
Ocean Science and Productivity Division
Science Branch
Pacific Biological Station
Nanaimo, B.C. V9R 5K6

PSARC Working Papers document the scientific basis for fisheries management advice in the Pacific Region. As such, they provide one component of the assessment process, and are not intended as comprehensive treatments of stock management.

Introduction

In 1958 a significant proportion of the Fraser River sockeye salmon spawning migration returned to the Fraser River via Johnstone Strait. This phenomenum hadn't occurred in the history of the IPSFC. Prior to 1958, the next most recent occurrence of this phenomenum was in 1936, just before the IPSFC was established. Since 1958, the interannual variation in the proportion of Fraser River sockeye salmon migrating through Johnstone Strait has attracted considerable scientific attention (Tully et al. 1960, Royal and Tully 1961, Wickett 1977, Hamilton 1985, Mysak 1986, Groot and Quinn 1987). One element common to all of these studies was the sockeye *northern diversion* estimates. These are estimates, produced annually, of the proportion of all Fraser River sockeye returning via Johnstone Strait (Fig. 1). The estimates were produced by the International Pacific Salmon Fisheries Commission for the years up to 1985, and during the past decade, by the Pacific Salmon Commission.

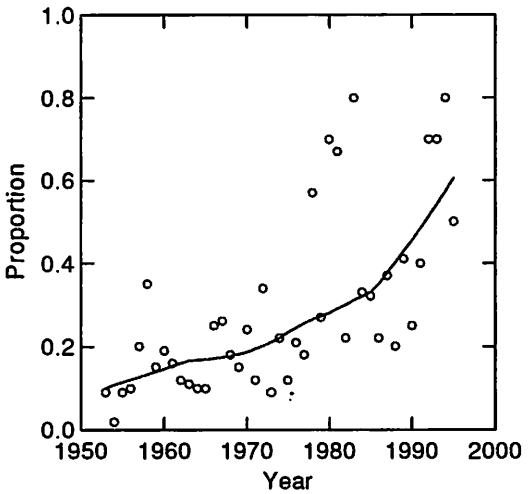


Fig. 1. Annual northern diversion estimates for Fraser River sockeye salmon

The methods used to compute the northern diversion estimates for sockeye have not been published and this has raised concerns about the reliability of the estimates. The Fraser River Sockeye Public Review Board recommended that additional research on northern diversion should be undertaken. The purpose of this document is to examine the northern diversion estimates produced by the two commissions in more detail and to assess their potential for continued use.

Methods

Fraser River sockeye that are caught either in Washington State, on the west coast of Vancouver Island, in Johnstone Strait, or in the Strait of Juan de Fuca are caught in fisheries approaching the Fraser River where the route taken by the sockeye around Vancouver Island is known with relatively little error. Those fish escaping from the approach route fisheries are not observed until they appear as catch at the rivermouth (DFO Statistical Area 29), as catch in upriver fisheries, or as spawning ground escapements. Escapements from the approach route fisheries provide no direct information about migration route without invoking the necessary assumptions to model the migration routes of sockeye that are not caught in the approach route fisheries.

Much of the historical data used in this report were obtained from Roos (1991) and the recent data were obtained from annual reports of the PSC and P. Ryall (pers. comm.). The total catch of Fraser River sockeye returning via the southern approach route was computed by summing the catches in DFO Statistical Areas 20, 21, 24-27 plus the U.S. catch of Fraser sockeye in Washington State. The sockeye catch in Area 23 (Barkley Sound) was omited because local stocks accounted for most of that sockeye catch. The total catch of Fraser River sockeye in the northern approach route was computed by summing catches in DFO Statistical Areas 11-16. I defined the sum of these northern plus southern catches as the *approach route catch.* That portion of the approach route catch taken annually in the northern route can be considered a *naive* northern diversion estimate if one assumes that the proportion of sockeye taken in the northern approach route by the fisheries in Johnstone Strait is the same as the proportion taken by the Juan de Fuca fisheries in the southern approach route. The consequences of violating this assumption will be examined.

naive estimate= northern approach catch / approach route catch

If every Fraser River sockeye was caught in the approach routes, the northern diversion would be known without error. That some fish escape the approach route fisheries is good for the fish but it complicates the process of estimating the northern diversion. Conversely, if no fish are caught in the approach route fisheries then the northern diversion estimate would be unreliable. This suggests that the greater the proportion of the total run that is taken in the approach route fisheries, the lower the uncertainty about the northern diversion estimate.

The northern diversion estimates produced by the commissions were compared with naive diversion estimates. Linear regression was used to determine that percentage of variance in the commissions' northern diversion estimates that can be explained by the naive diversion estimate. I hypothesized that if the intercept passed through the origin with a slope equal to 1.0 then the current northern diversion estimates are based almost exclusively on approach route catches. The degree to which the commissions' northern diversion estimates deviate from the fitted regression line is a measure of the degree to which the northern diversion estimate is determined by factors other than catch.

If most of the signal in the northern diversion estimate is generated from the approach route catches, it follows that the escapement from the approach routes must be of little consequence in the process of generating the northern diversion estimate. This raised the question 'How wrong would the northern diversion estimates be if the assumptions about harvest proportions are wrong?' This line of reasoning led to the development of a simple simulation to examine the question posed above.

Assume an arbitrary northern diversion estimate, develop a daily run timing curve (Schnute and Sibert 1983) and allocate fish to the northern or southern migration routes using a known northern diversion estimate. Assume that fishing will occur on two of seven days per week and that these will be chosen randomly. Establish simulated fisheries in the two approach routes that remove fish at a known rate in each fishery. Knowing the true harvest rates and catches in each approach route, compute the true simulated diversion estimate. Now introduce random error in knowledge of the harvest proportions and measure its influence on the diversion estimate. The simulations were run with northern diversion estimates varied through 20, 40, 60, and 80% with harvest proportions in both the southern and northern fisheries varied through 20, 40, 60, and 80%. Random harvest errors of ±20% were added to the true harvest proportions in both fisheries. One hundred iterations were run at each combination of diversion and harvest levels.

Results

In the 44 years between 1953 and 1995, the percentage of the total run observed in the approach route fisheries varied from 28.1% (1995) to 71.6% (1985) with a mean of 56.6% (Fig. 2). There is no significant trend in this percentage over time.

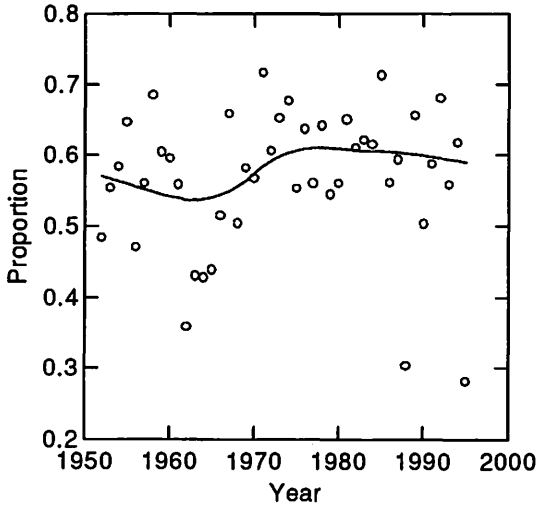


Fig. 2. Proportion of Fraser River sockeye caught in approach route fisheries.

The naive diversion estimates follow a pattern that is very similar to the northern diversion estimates (Fig. 3).

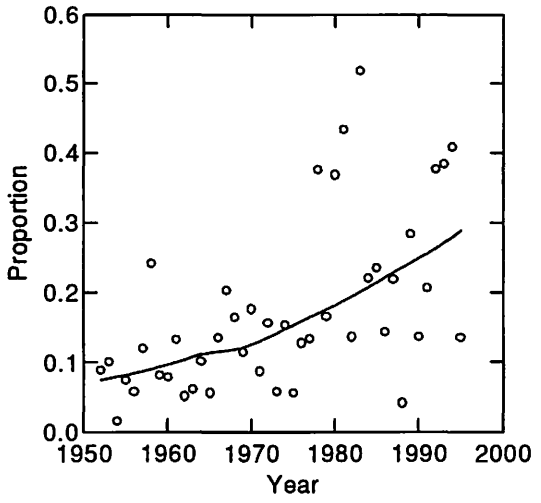


Fig. 3. Proportion of the total Fraser sockeye run caught in Johnstone Strait

A major finding of this study is that the naive diversion estimate accounts for 92.6% to 93.7% (depending on whether the data from the 2 commissions is combined or not) of the variation in the northern diversion estimates (Fig. 4).

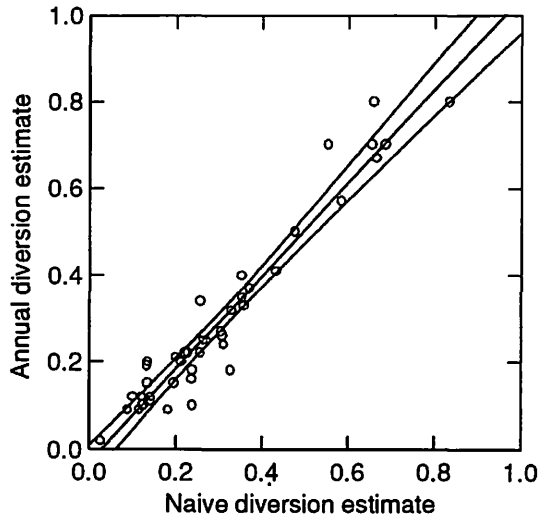


Fig. 4. Regression of salmon commission estimates versus naive diversion estimates derived from catch data only.

The model passes through the origin with a slope that is not significantly different from one i.e. northern diversion estimate= naive estimate + error. This implies that the northern diversion estimate is 0% if there is no catch in the northern approach, or 100% if there is no catch via the southern approach. The remaining 7% of the variation in the northern diversion estimates that is not accounted for by the naive estimate is likely the result of factors other than catch (eg. effort). It may also be the result of revisions to the catch data over the years, or due to small differences between the way I assigned catches to approach routes and how the commissions may have done them.

The residuals from the model (Fig. 5) indicate those years that received the greatest adjustments from the naive estimate.

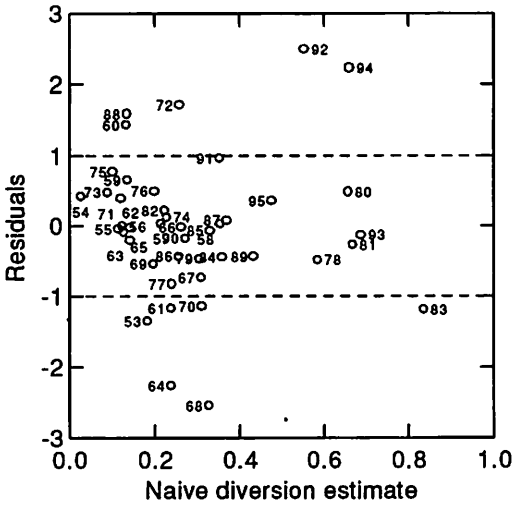


Fig. 5. Standardized residuals from Fig. 4 indicating those years where the commissions' estimates differ from those obtained from catch data only.

The years 1960, 1972, 1988, 1992, 1994 have the greatest (>1 s.d. from the mean) positive deviations from the model. In these years, the commissions computed a greater northern diversion than would have been determined from the catches alone. In 1953, 1961, 1964, 1968, 1970, and 1983, the reverse. The residuals on the 1992 cycle are the greatest (Fig. 6) and those on the 1994 cycle line are generally the lowest. The 1992 cycle year has the smallest average return and the 1994 cycle has the largest total return.

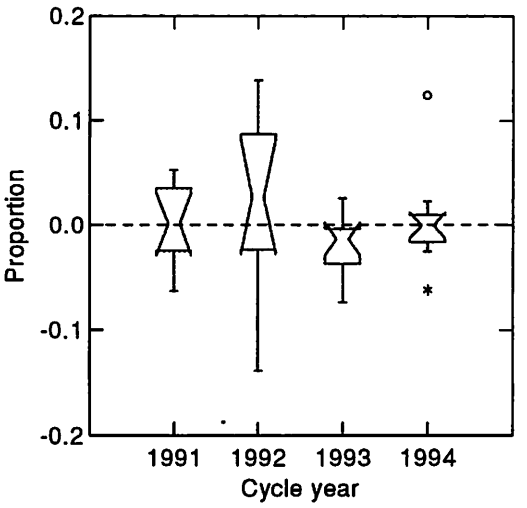


Fig. 6. Distribution of residuals from Fig. 4 by cycle year.

The run timing curve used in the simulations appears in Figure 7.

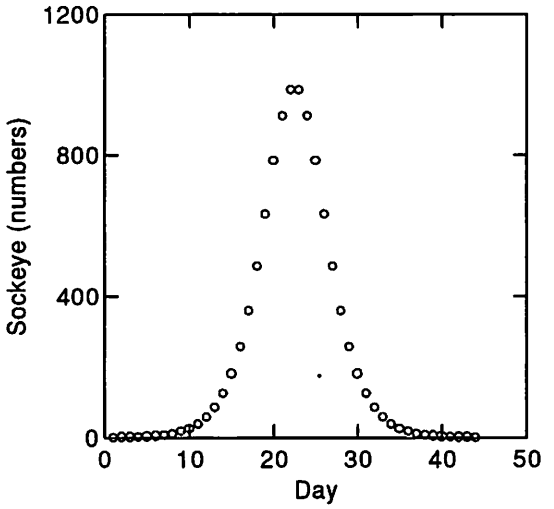


Fig. 7. Simulated daily run timing curve for sockeye approaching the northern end of Vancouver Island.

The simulation results indicated that the naive diversion estimates are not unreasonable approximations of the true diversion (Fig. 8). The bias contours (naive estimate - true value) tended toward values less than ±0.10 through a wide range of simulated harvest proportions and northern diversions. The bias tended to zero when the proportions harvested in the northern and southern approach routes were equal, whether the diversion was high or low. As northern approach route harvest proportions exceeded those in the southern approach route, the naive diversion estimate overestimated the true value. The naive diversion estimate underestimated the true value when southern approach

route harvest proportions exceeded those in the northern approach route. There were asymmetries in the bias response when the diversion rate was known to be low or high. Note in Figure 8 that when the northern diversion was 0.2, the +0.1 bias contour appeared but the -0.1 contour is out of range. Note also that the reverse occurs when the northern diversion is 0.8. At low northern diversion values, the magnitude of the bias increased more rapidly at high northern and low southern harvest proportions than at high southern and low northern harvest proportions. The reverse occurred when the true northern diversion was high.

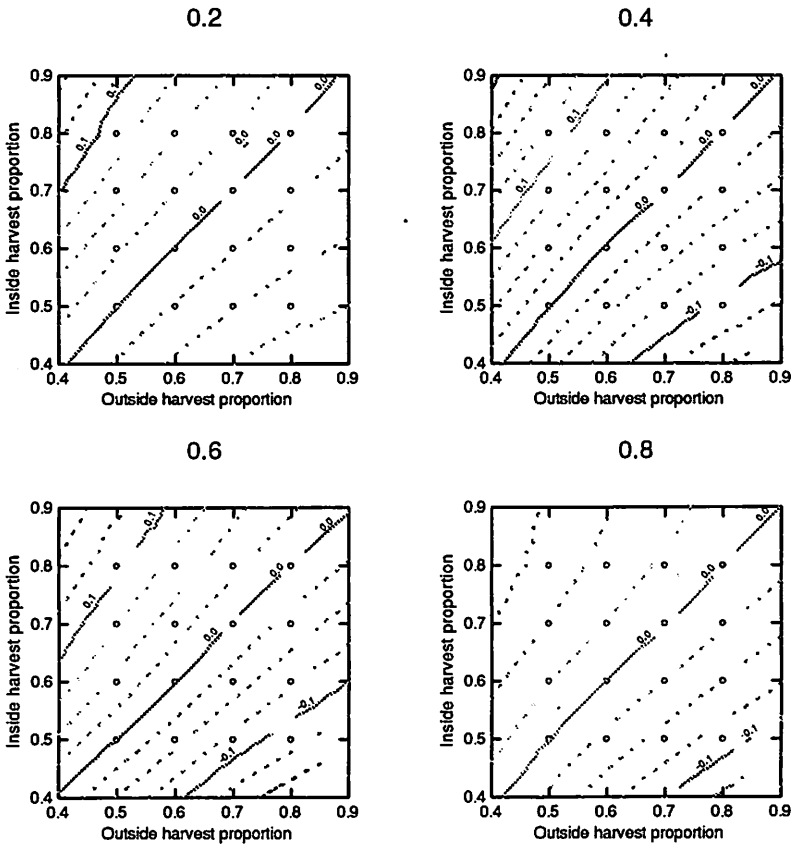


Fig. 8. Bias (naive-true diversion) contours at various levels of northern and southern harvest. Each panel portrays the bias at a fixed northern diversion (indicated above each panel).

Figure 8 indicated the tendencies after repeated Monte Carlo trials. In fact, the variation at each combination of northern diversion, northern and southern approach harvest proportions will be greater than that indicated. The range of uncertainty for any individual trial exceeds these values. Figure 9A shows the variation of individual trials at various levels of northern diversion with harvest proportions in both routes set at 0.8. As expected, the bias is zero but, encouragingly, the variation of individual points about the median is low. Similar ranges of variation are apparent at other combinations of harvest proportions in the two approach routes.

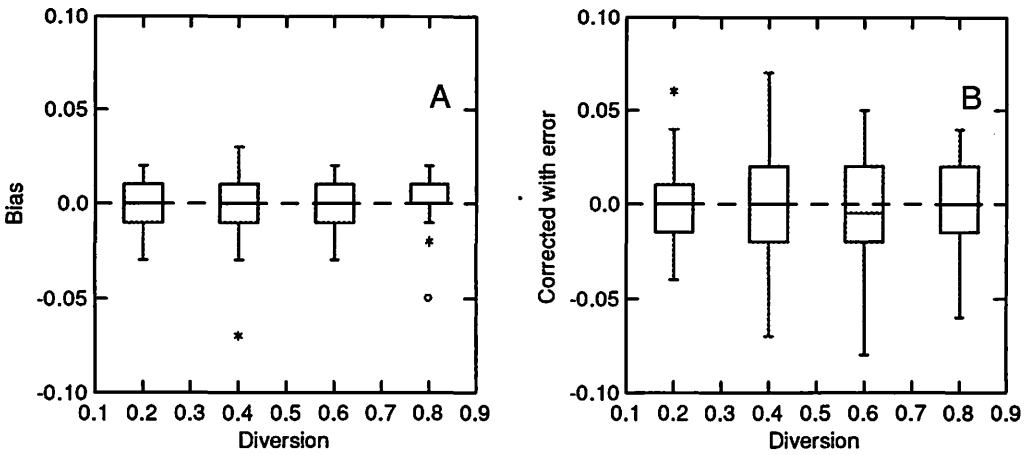


Fig. 9. (A) Distribution of bias (naive-true northern diversion estimates) at various northern diverson levels with harvest proportions in northern and southern approach fisheries fixed at -0.8, (B) corrected with errors of up to 20% applied to the true harvest proportions.

If the harvest rates are known without error in each route, the catches can be accurately expanded to the total runs through each approach route. Figure 9B shows the effect of applying inaccurate harvest rates on the annual estimates. With random harvest rate errors (drawn from a uniform distribution) of up to ±20%, the variability of individual estimates is increased beyond what would be obtained if the uncorrected naive estimates were used.

Discussion

The data and simulations suggest that the commissions' northern diversion estimates were providing the major interannual signal in the true northern diversion. In part, this is because the recent frequency of high northern diversions since 1976 has provided significant contrast in the data. Prior to 1977, there are periods of lesser contrast in the northern diversion series where errors resulting from poor estimates of harvest rate could have introduced errors in the northern diversion that exceeded the interannual variation.

How reliable are these estimates prior to 1976? Most researchers of this topic have hypothesized that a physical process determines the migration route. That Wickett (1977) was able to model northern diversions prior to 1977 using a variety of physical correlates including Fraser River discharge suggests that even during those periods of lower variation, the commissions produced northern diversion estimates that may have been reasonably well determined. This assumes that the commissions did not use these same physical variables to develop their estimates.

There are a number of potential errors in this analysis. Firstly, all Washington State catch was attributed to the southern route. In some years, there may be unknown numbers of sockeye approaching the rivermouth via the northern route and straying into US waters prior to entering the river. Area 24-27 sockeye catches have been attributed to the southern route. In fact, they may not even be Fraser river sockeye, however, the numbers caught in these areas is so small when compared to other locations that the outcome will not be greatly affected by including or excluding these catches.

That the deviations of the northern diversion estimates from the naive estimates are greatest on the 1992 cycle suggests that the commissions have traditionally been inclined to make greater adjustments in years when the total run is the lowest. A recent outlier on the 1994 cycle (1994) suggests that this pattern may be breaking down. As the management of the Canadian sockeye fishery has changed dramatically (eg. area licensing, fleet reduction) in recent years, the application of the conclusions of this work to future northern diversion estimates may not hold, particularly if effort is sufficiently reduced to effect the catch.

The simulation exercise did not attempt to model the complexity of fishery and migration of sockeye salmon, because it was not necessary. Perhaps even the simulations were not necessary because the northern diversion is such a simple process. Only 2 routes are possible; removals happen along the routes but the rate and timing of the removals is not relevant to the spatial and temporal scales of the northern diversion; and the escapement from the approach route fisheries is not relevant to the northern diversion without invoking assumptions that can't be tested.

Recommendations

1. The PSC should be encouraged to document their method of estimating the northern diversion.

2. The historical time series of northern diversion estimates should continue to be used by the Department for forecasting sockeye migration routes around Vancouver Island.

References

Groot, C. and Quinn, T.P. 1987. Homing migration of sockeye salmon, *Oncorhynchus nerka*, to the Fraser River. Fish. Bull. 85:455-469.

Hamilton, K. 1985. A study of the variability of the return migration route of the Fraser River sockeye salmon (*Oncorhynchus nerka*). Can. J. Zool. 63:1930-1943.

Mysak, L.A. 1986. El Niño, interannual variability and fisheries in the northeast Pacific Ocean. Can. J. Fish. Aquat. Sci. 43:464-497.

Roos, J.F. 1991. Restoring Fraser River salmon: A history of the International Pacific Salmon Fisheries Commission 1937-1985.

Royal, L.A. and Tully, J.P. 1961. Relationship of variable oceanographic factors to migrations and survival of Fraser River salmon. Calif. Coop. Fish. Invest. 8:65-68.

Schnute, J. and Sibert, J. 1983. The salmon terminal fishery: a practical, comprehensive model. Can. J. Fish. Aquat. Sci.:40:835-853.

Tully, J.P., Dodimead, A.J. and Tabata, S. (1960) An anomalous increase in temperature in the ocean off the Pacific coast of Canada through 1957 and 1958. *J. Fish. Res. Bd. Can.* 17:61-80.

Wickett, W.P. 1977. Relationship of coastal oceanographic factors to the migration of Fraser River sockeye salmon (*Oncorhynchus nerka* W.). ICES CM 1977/M:26.

Table 1. Fraser sockeye data used to assess northern diversion estimates from 1953-1995). Sockeye are tabulated as numbers. Northern diversion is reported as a proportion as are the approach route catches (number of sockeye observed in areas where the migration route is known).

YEAR	N. Divert. (prop.)	N. Catch (no.)	S. Catch (no.)	US Catch (no.)	Fraser Catch (no.)	Escape ment (no.)	Native Catch (no.)	Apprch. Catch (prop.)	Total Run (no.)
1953	0.09	616	741	2032	1328	1288	113	0.55	6119
1954	0.02	174	1618	4806	2163	2448	95	0.58	11304
1955	0.09	206	577	1007	528	383	66	0.65	2767
1956	0.10	166	279	907	582	873	65	0.47	2872
1957	0.20	655	720	1689	645	1662	97	0.56	5467
1958	0.35	4427	2834	5257	1809	3867	83	0.68	18276
1959	0.15	395	709	1811	898	949	65	0.60	4826
1960	0.19	275	598	1199	697	628	84	0.60	3481
1961	0.16	631	645	1378	716	1250	137	0.56	4757
1962	0.12	182	343	759	537	1624	137	0.36	3582
1963	0.11	246	172	1314	496	1607	190	0.43	4025
1964	0.10	184	82	508	464	426	146	0.43	1810
1965	0.10	181	227	1026	864	852	121	0.44	3272
1966	0.25	753	776	1337	629	1920	154	0.51	5569
1967	0.26	1395	1037	2088	878	1353	107	0.66	6858
1968	0.18	525	197	886	834	627	124	0.50	3193
1969	0.15	580	790	1586	957	1007	159	0.58	5079
1970	0.24	1089	1065	1350	575	1943	151	0.57	6172
1971	0.12	678	2109	2816	1317	748	154	0.72	7821
1972	0.34	592	573	1128	527	830	133	0.61	3783
1973	0.09	401	1492	2628	1064	1181	164	0.65	6929
1974	0.22	1283	1889	2462	714	1758	222	0.68	8328
1975	0.12	210	293	1564	427	993	253	0.55	3740
1976	0.21	564	943	1322	558	823	233	0.64	4443
1977	0.18	769	661	1789	1176	1113	244	0.56	5753
1978	0.57	3497	1121	1362	581	2514	238	0.64	9313
1979	0.27	1017	549	1776	1098	1408	292	0.54	6141
1980	0.70	1119	119	466	302	849	186	0.56	3042
1981	0.67	3291	350	1294	773	1443	441	0.65	7591
1982	0.22	1820	3484	2867	756	4024	430	0.61	13380
1983	0.80	2591	45	469	531	997	362	0.62	4995
1984	0.33	1248	594	1635	888	932	356	0.62	5654
1985	0.32	3176	3371	3063	1314	2139	424	0.71	13487
1986	0.22	2261	3721	2853	2668	3717	534	0.56	15754
1987	0.37	1617	852	1905	612	1920	468	0.59	7374
1988	0.20	151	321	651	740	1418	416	0.30	3696
1989	0.41	5049	4171	2440	2435	3108	572	0.66	17775
1990	0.25	2788	5075	2427	3279	6085	808	0.50	20462
1991	0.40	2432	2584	1882	898	3341	607	0.59	11744
1992	0.70	2073	972	700	270	1120	368	0.68	5503
1993	0.70	8700	1076	2876	2776	6372	848	0.56	22647
1994	0.80	6085	1074	2055	1642	3118	928	0.62	14902
1995	0.50	508	131	415	202	1751	743	0.28	3742